29/7/26

Activity -02:-

stack, Queue, Linked List:-

①

① stack operations

- push (54) → stack: [54]
- push (52) → stack: [54, 52]
- pop () → removes 52, stack: [54]
- push (55) → stack: [54, 55]
- push (62) → stack: [54, 55, 62]
- s=pop () → removes 62, so s=62

Queue Operations:

- Enqueue (21) → queue: [21]
- enqueue (24) → queue: [21, 24]
- dequeue () → removes 21, queue: [24]
- enqueue (28) → queue [24, 28]
- enqueue (32) → queue [24, 28, 32]
- q = dequeue () → removes 24, so q=24

s+q = 62 + 24

=86

② postfix Expression

Expression : 10.5+60.6/*8−

- 10+5 = 15
- 60/6 = 10
- 15×10 = 150
- 150 − 8 = 142

Answer = 142

③ Two Queue

minimum number of Enqueue operations

on . $Q_1 = 6$

Answer = 6

④ Reverse operation on stack

→ Using REVERSE changes the stack order in

$\Theta(n)$ time

⑤ post fix expression

Expression : 823^/23*+51*−

After the first * is evaluated, the top two

stack elements are:

Answer: 6, 5

⑥ Infix to postfix

Infix: a+ b×c -d ^e ^f

postfix : abc + def ^^ -*

⑦ stack operations

Operations:

* push (10)
* PUSH (20)
* pop → 20
* PUSH (10)
* PUSH (20)
* pop → 20
* pop → 10
* pop → 10
* PUSH (20)
* pop → 20

Popped sequence: 20, 20, 10, 10, 20